

Homework (Jalapeno)

- Q1.** Solve for x : $|x| = 5$
(A) $x = 5$
(B) $x = -5$
(C) $x = 5$ or $x = -5$
(D) $x = 0$
(E) $x = 10$
- Q2.** The ancient Egyptians built the Great Pyramid with a base perimeter of $|x + 10| = 20$ meters. What could x be?
(A) $x = 10$
(B) $x = -10$
(C) $x = 10$ or $x = -30$
(D) $x = 0$
(E) $x = 5$
- Q3.** Solve the equation $|2x - 3| = 7$
(A) $x = 2$ or $x = 5$
(B) $x = -2$ or $x = -5$
(C) $x = 2$ or $x = -5$
(D) $x = -2$ or $x = 5$
(E) $x = 0$
- Q4.** Archaeologists discovered an ancient artifact with an inscription indicating a distance of $|x - 2| = 9$ miles. What are the possible values for x ?
(A) $x = 11$
(B) $x = -7$
(C) $x = 11$ or $x = -7$
(D) $x = 0$
(E) $x = 5$
- Q5.** The absolute value equation $|x + 4| = 2$ has solutions of
(A) $x = -2$
(B) $x = -6$
(C) $x = -2$ or $x = -6$
(D) $x = 0$
(E) $x = 2$
- Q6.** An ancient Greek mathematician solved the equation $|3x - 1| = 11$. What are the possible values for x ?
(A) $x = 4$
(B) $x = -4$
(C) $x = 4$ or $x = -10/3$
(D) $x = 0$
(E) $x = 2$
- Q7.** Solve the equation $|x/2 + 1| = 3$
(A) $x = 4$
(B) $x = -8$
(C) $x = 4$ or $x = -8$
(D) $x = 0$
(E) $x = 2$
- Q8.** Historians found an ancient scroll indicating the height of a building to be $|x - 5| = 3$ meters. What are the possible values for x ?
(A) $x = 8$
(B) $x = 2$
(C) $x = 8$ or $x = 2$
(D) $x = 0$
(E) $x = 5$

Open Ended Questions (FRQ)

- Q9.** Solve the equation $|2x + 5| = |x - 3|$ and show all steps. Check your solutions for extraneous results.
- Q10.** The equation $|x - 2| = |3x + 1|$ represents the distance between two ancient monuments. Solve the equation and explain the meaning of your solutions in the context of the problem.

Chef's Tips

- Check for extraneous solutions
- Use the definition of absolute value to solve equations
- Consider the historical context of the problems